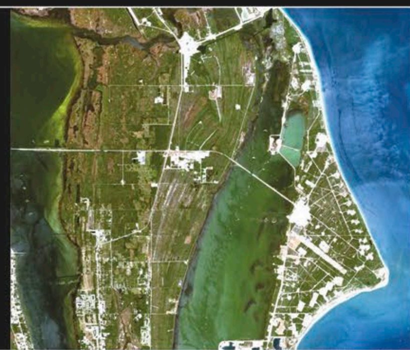
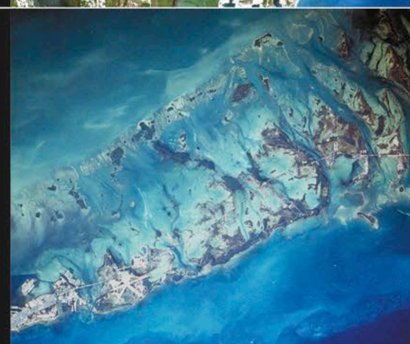
A SHORT TOUR OF THE UNIVERSE



KENNEDY SPACE CENTER
Many of humankind's first ventures into space were set in motion on the launchpads of Kennedy Space Center. This remains the busiest launch and landing site of the US space program, and it was also the main base for the Space Shuttle.



THE FLORIDA COAST
The islands and reefs of the Florida Keys are seen here from Earth orbit. The reefs are partly made of living organisms in the form of corals. To date, life has not been found anywhere other than on Earth, but the search for alien life will be perhaps the most fascinating quest of the 21st century.

THE NIGHT SKY has always evoked mystery and wonder. Since antiquity, astronomers have tried to understand the patterns of the "fixed stars" and the motions of the Moon and planets. The motive was partly a practical one, but there has always been a more "poetic" motivation, too—a quest to understand our place in nature. Modern science reveals a cosmos far vaster and more varied than our ancestors could have envisioned.

No continents on Earth remain to be discovered. The exploratory challenge has now broadened to the cosmos. Humans have walked on the Moon; uncrewed spacecraft have beamed back views of all the planets; and some people now living may one day walk on Mars.

The stars, fixed in the "vault of heaven," were a mystery to the ancients. They are still unattainably remote, but we know that many of them are shining even more brightly than the Sun. Within the last decade, we have learned something remarkable that was long suspected: many stars are, like our Sun, encircled by orbiting planets. The number of known planetary systems already runs into hundreds—there could, all together, be a billion in our galaxy. Could some of these planets resemble the Earth and harbor life? Even intelligent life?

All the stars visible to the unaided eye are part of our home galaxy—a structure so vast that light takes a hundred thousand years to cross it. But this galaxy, the Milky Way, is just one of billions visible through large telescopes. These galaxies are hurtling away from each other, as though they had all originated in a "big bang" 13 or 14 billion years ago. But we don't know what banged, nor why it banged.

The beauty of the night sky is a common experience of people from all cultures; indeed, it is something that we share with all generations since prehistoric times. Our modern perception of the "cosmic environment" is even grander. Astronomers are now setting Earth in a cosmic context. They seek to understand how the cosmos developed its intricate complexity—how the first galaxies, stars, and planets formed and how, on at least one planet, atoms assembled into creatures able to ponder their origins. This book sets humanity's concept of the cosmos in its historic context and presents the latest discoveries and theories. It is a beautiful "field guide" to our cosmic habitat: it should enlighten and delight anyone who has looked up at the stars with wonder and wished to understand them better.

MARTIN REES

THE MOON
1.3 light-seconds from Earth
The Earth is seen here rising above the horizon of its satellite, the Moon. Our home planet's delicate biosphere contrasts with the sterile moonscape on which the Apollo astronauts left their footprints.